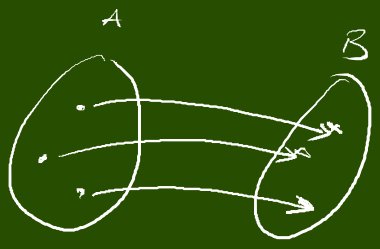
$k = ?$ $V = ?$ bază unică

A mulțime numărabilă $\iff \exists f: \mathbb{N} \longrightarrow A$ bij

$A = \{f(0), f(1), \dots\}$



$A \sim B$ echipotente dacă $\exists f: A \to B$ bijectivă.

V sp. vect. B_1, B_2 baze în V

$\Rightarrow \exists f: B_1 \longrightarrow B_2$ bijectivă.

$\dim_K V = n$

$v_1 \dots v_n \in V$ liniar independenți $\Rightarrow$ bază.

$\Leftarrow \mathrm{Sp}\{v_1 \dots v_n\} \subseteq V$

$\subseteq$ mereu

$\supseteq$?

pp. că există $v \in V$, $v \notin \mathrm{Sp}\{v_1 \dots v_n\}$.

$v_1 \dots v_n, v$ liniar indep? NU

$n+1$ vectori

$\Rightarrow \exists \lambda_1 \dots \lambda_n, \lambda \in K$ nu toți nuli a.î.:

$\lambda_1 v_1 + \dots + \lambda_n v_n + \lambda v = 0 \mid \cdot \lambda^{-1}$

$v_1 \dots v_n$ liniar indep.

$\lambda = 0 \Rightarrow \lambda_1 v_1 + \dots + \lambda_n v_n = 0 \Rightarrow \lambda_1 = \dots = \lambda_n = 0$

fals.

$\lambda \neq 0 \Rightarrow v = -\lambda_1 \lambda^{-1} v_1 - \dots - \lambda_n \lambda^{-1} v_n$

$v \in \mathrm{Sp}\{v_1 \dots v_n\}$ fals.

$\Rightarrow$ presupunerea făcută e falsă $\Rightarrow$

$V = \mathrm{Sp}\{v_1 \dots v_n\} \Rightarrow \{v_1 \dots v_n\}$ bază.

Exemplu. $V = \mathbb{R}_2[X]$ $\quad K = \mathbb{R}$

$\dim_{\mathbb{R}} V = 3$ $\quad \{1, X, X^2\}$

a) $P_1, P_2 \in \mathbb{R}_2[X]$ $\quad P_1(X) = 1 - X,\ P_2(X) = 1 + X^2$

$\lambda_1 P_1(X) + \lambda_2 P_2(X) = 0$

$P_1(X) = -\lambda_2 \lambda_1^{-1} P_2(X)$

$\Rightarrow P_1, P_2$ liniar indep.

Caut $P_3 \in \mathbb{R}_2[X]$ a.i. P_1, P_2, P_3 lin. indep.

($\Leftrightarrow$ bază).

$P_3(X) = a + bX + cX^2$

$\lambda_1 P_1 + \lambda_2 P_2 + \lambda_3 P_3 = 0 \quad \Rightarrow \lambda_1 = \lambda_2 = \lambda_3 = 0$

$$\begin{cases} \lambda_1 + \lambda_2 + \lambda_3 a = 0 \\ -\lambda_1 \quad + \lambda_3 b = 0 \\ \quad \lambda_2 + \lambda_3 c = 0 \end{cases} \qquad \underbrace{\begin{pmatrix} 1 & 1 & a \\ -1 & 0 & b \\ 0 & 1 & c \end{pmatrix}}_{A}$$

Alegem a, b, c a.i. $\det A \neq 0$

$$\det A = -a + c - b \neq 0$$

$$c \neq a + b$$

de exemplu $a = b = 0$, $c = 1$ $(P_3(X) = X^2)$

$\Rightarrow P_1, P_2, P_3$ lin. indep $\Rightarrow$ bază în $\mathbb{R}_2[X]$

---

b) $\cos nx = \ldots (\cos x)$

$n \in \mathbb{N}$

$\cos 0x = 1$ $\quad T_0(X) = 1$

$\cos 1x = \cos x$ $\quad T_1(X) = X$

$\cos 2x = 2\cos^2 x - 1$ $\quad T_2(X) = 2X^2 - 1$

$\cos 3x = 4\cos^3 x - 3\cos x$ $\quad T_3(X) = 4X^3 - 3X$

$\cos(n+1)x = \cos(nx + x) \longrightarrow T_{n+1}(X)$? $T_n(X)$

polinoame Cebîşev

Deci $\mathbb{R}_3[X]$ T_0, T_1, T_2, T_3 bază.

c) temă! $R_2[X] = Sp\{Q_1, Q_2, Q_3, Q_4\}$

$Q_1(X) = 1+X$, $Q_2(X) = 2-X+X^2$, $Q_3 = 3X-2X^2$,

$Q_4(X) = -1+5X+X^2$

Q_4 - poate fi eliminat ? Da, dacă

$Q_4 \in Sp\{Q_1, Q_2, Q_3\}$?

$\Leftrightarrow \exists \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ a.i. $Q_4 = \lambda_1 Q_1 + \lambda_2 Q_2 + \lambda_3 Q_3$

$$\begin{cases} \lambda_1 + 2\lambda_2 = -1 \\ \lambda_1 - \lambda_2 + 3\lambda_3 = 5 \\ \lambda_2 - 2\lambda_3 = 1 \end{cases}$$

$$\Rightarrow \left(\begin{array}{ccc|c} \boxed{1} & 2 & 0 & -1 \\ \langle 1 \rangle & -1 & 3 & 5 \\ 0 & 1 & -2 & 1 \end{array}\right) \xrightarrow{L_2 - L_1} \left(\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & -3 & 3 & 6 \\ 0 & 1 & -2 & 1 \end{array}\right) \xrightarrow{L_2 : (-3)}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & \boxed{-1} & 1 & 2 \\ 0 & \langle 1 \rangle & -2 & 1 \end{array}\right) \xrightarrow{L_3 + L_2} \left(\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & -1 & 3 \end{array}\right)$$

$$\begin{cases} \lambda_1 + 2\lambda_2 = -1 \\ -\lambda_2 + \lambda_3 = 2 \\ -\lambda_3 = 3 \end{cases} \Rightarrow \begin{array}{l} \lambda_1 = 9 \\ \lambda_2 = -5 \\ \lambda_3 = -3 \end{array}$$

$\Rightarrow \mathbb{R}_2[x] = Sp\{Q_1, Q_2, Q_3\}$ $\Rightarrow$

$\dim_{\mathbb{R}} \mathbb{R}_2[x] = 3$

$\Rightarrow \{Q_1, Q_2, Q_3\}$ base.

$U \subseteq V$

$\dim_{\mathbb{K}} V = \infty \qquad \mathbb{K} = \mathbb{R} \qquad V = \mathbb{R}[x]$

$U = \{ P \in \mathbb{R}[x] \mid P(x) = P(-x) \}$

polin. pare

$P(x) = a_0 + a_2 x^2 + a_4 x^4 + \dots + a_{2n} x^{2n}$

U Subsp. in $\mathbb{R}[x]$

$\dim_{\mathbb{R}} U = \infty \qquad U = Sp\{1, x^2, x^4, \dots, x^{2n}, \dots\}$

$\Rightarrow$ $U = V$? NU

- finit $B \subseteq A$ $\operatorname{card} B = \operatorname{card} A \Rightarrow$

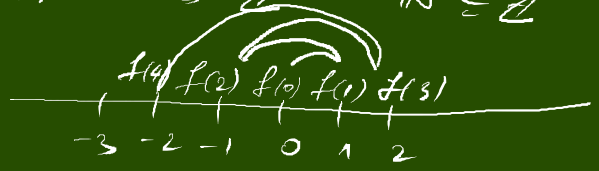
$B = A$

- infinit $B \subsetneq A$ $\exists f : B \to A$ bijectivă

$\Rightarrow B = A$?

f

$\mathbb{N} \longrightarrow \mathbb{Z}$ $\mathbb{N} \subsetneq \mathbb{Z}$ $\mathbb{N} \neq \mathbb{Z}$



$$f(n) = \begin{cases} 0, & n = 0 \\ k+1, & n = 2k+1 \\ -k, & n = 2k \end{cases}$$

$U \subseteq V$ subspații

$\dim_k V = n \Rightarrow 0 \leq \dim_k U \leq n$

$\dim_k U = n \Rightarrow U = V$

$\dim_k U = 0 \Rightarrow U = \{0\}$

---

k^n < $\ker(A)$, $\mathrm{Im}(A)$ > $\dim$?

Exemple $A = \left(\begin{array}{ccc|c} 1 & 0 & -1 & \\ 0 & 2 & 2 & \\ \hline -1 & 1 & 2 & \\ 2 & 3 & 1 & \end{array}\right) \in \mathcal{M}_{4,3}(\mathbb{R})$

$C_4^3 = 4$ mineurs 3×3

$\mathrm{rang}(A) = 2$

$\begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} \neq 0$

$\begin{vmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ -1 & 1 & 2 \end{vmatrix} = 0$ ✓

$\begin{vmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ 2 & 3 & 1 \end{vmatrix} = 0$ ✓

$\Rightarrow \dim_{\mathbb{R}} \mathrm{Im}(A) = 2$, $\mathrm{Im}(A) \subseteq \mathbb{R}^4$

$\dim_{\mathbb{R}} \ker(A) = 3 - \mathrm{rang}(A) = 1$, $\ker(A) \subseteq \mathbb{R}^3$

$$\ker(A) = \{ X \in \mathbb{R}^3 \mid AX = O_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad (4,3)(3,1)$$

$$\begin{cases} x - z = 0 \\ 2y + 2z = 0 \\ -x + y + 2z = 0 \\ 2x + 3y + z = 0 \end{cases} \Rightarrow \begin{matrix} x = z \\ y = -z \end{matrix}$$

$$\ker(A) = \left\{ \begin{pmatrix} z \\ -z \\ z \end{pmatrix} \mid z \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

$\Rightarrow \left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$ baza în $\ker(A)$

$\dim_{\mathbb{R}} \ker(A) = 1$

$$\mathrm{Im}(A) = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 0 \\ 2 \\ 1 \\ 3 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} -1 \\ 2 \\ 2 \\ 1 \end{pmatrix}}_{v_3} \right\} \subseteq \mathbb{R}^4$$

$\dim_{\mathbb{R}} \mathrm{Im}(A) \leq 3$

v_1, v_2, v_3 lin. indep. $\Leftrightarrow \operatorname{rang}(\overbrace{v_1 | v_2 | v_3}^{A}) = 3$

v_1, v_2 lin. indep. ? $\Leftrightarrow \operatorname{rang}(v_1 | v_2) = 2$ ✓

$v_3 \in \mathrm{Sp}\{v_1, v_2\}$?

$v_3 = (-1)v_1 + 1 \cdot v_2$ $\quad \exists \lambda_1, \lambda_2 \in \mathbb{R}$ s.t.

$v_3 = \lambda_1 v_1 + \lambda_2 v_2$

$$\begin{cases} \lambda_1 = -1 \\ 2\lambda_2 = 2 \\ -\lambda_1 + \lambda_2 = 2 \\ 2\lambda_1 + 3\lambda_2 = 1 \end{cases} \Rightarrow \begin{matrix} \lambda_1 = -1 \\ \lambda_2 = 1 \end{matrix}$$

$\Rightarrow \mathrm{Im}(A) = \mathrm{Sp}\{v_1, v_2, v_3\} = \mathrm{Sp}\{\underbrace{v_1, v_2}_{\text{lin. indep.}}\}$

$\Rightarrow \{v_1, v_2\}$ bază în $\mathrm{Im}(A) \Rightarrow \dim_{\mathbb{Q}} \mathrm{Im}(A) = 2$

$= \mathrm{rang}(A)$

K V $\dim_K V = n$

$U, W \subseteq V$ subspații

$\dim_K U$ $\dim_K W$

$\dim_K (U+W)$

$\dim_K (U \cap W)$

A, B

$A \cup B$

$A \cap B$

$\mathrm{card}(A \cup B) = \mathrm{card}(A) + \mathrm{card}(B) - \mathrm{card}(A \cap B)$

Pas 1 bază în $U \cap W$ B

Pas 2 extindem B la o bază în U

$B \cup B'$ bază în U

extindem $\mathcal{B}$ la o bază în W

$\mathcal{B} \cup \mathcal{B}''$

Pas 3 $\mathcal{B} \cup \mathcal{B}' \cup \mathcal{B}''$ bază în $U + W$

$\left\{ \begin{array}{l} \text{lin. indep.} \\ \text{sist. de gen.} \end{array} \right.$

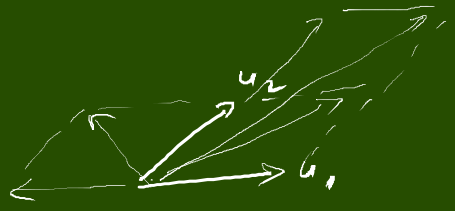
Exemplu. $\mathbb{R}^3$ $U = Sp\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$ plan

$W = Sp\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ plan

bază, dim pt U, W, $U \cap W$, $U + W$.

$U = Sp\{u_1, u_2\} \Rightarrow$

$\dim_{\mathbb{R}} U \leq 2$



$U \subseteq \mathbb{R}^3 \Rightarrow \dim_{\mathbb{R}} U \leq 3$

u_1, u_2 lin. indep $/\Rightarrow$

$U = Sp\{u_1, u_2\}$

$\Rightarrow \{u_1, u_2\}$ bază $\Rightarrow \dim_{\mathbb{R}} U = 2$

analog $\{w_1, w_2\}$ bază $\Rightarrow \dim_{\mathbb{R}} W = 2$

Aplicăm eliminarea Gauss pt. det. dim. lui $U+W$, $U \cap W$

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 1 & 2 & 2 & u_2 \\ 1 & 1 & 0 & w_1 \\ 0 & 1 & 1 & w_2 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & \textcircled{2} & 0 & u_2 - u_1 \\ 0 & 1 & -2 & w_1 - u_1 \\ 0 & 1 & 1 & w_2 \end{array}\right)$$

$$\xrightarrow{L_2 \cdot \frac{1}{2}} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & \textcircled{1} & 0 & (u_2 - u_1)/2 \\ 0 & \langle 1 \rangle & -2 & w_1 - u_1 \\ 0 & \langle 1 \rangle & 1 & w_2 \end{array}\right)$$

$$\begin{array}{l} L_3 - L_2 \\ \longrightarrow \\ L_4 - L_2 \end{array} \left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & 1 & 0 & (u_2 - u_1)/2 \\ 0 & 0 & -2 & w_1 - u_1 - (u_2 - u_1)/2 \\ 0 & 0 & 1 & w_2 - (u_2 - u_1)/2 \end{array}\right)$$

$U + W \Gamma$

$L_3 \leftrightarrow L_4$

$L_4 + 2L_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & u_1 \\ 0 & 1 & 0 & (u_2 - u_1)/2 \\ 0 & 0 & 1 & w_2 - \frac{u_2}{2} + \frac{u_1}{2} \\ 0 & 0 & 0 & w_1 - u_1 - \frac{u_2}{2} + \frac{u_1}{2} + 2w_2 \end{array}\right)$$

$U \cap W$

$$- u_2 + u_1$$

$$\Rightarrow \quad w_1 + 2w_2 - \frac{3u_2}{2} + \frac{u_1}{2} = 0$$

$$\underbrace{w_1 + 2w_2}_{\in W} = \underbrace{\frac{3u_2}{2} - \frac{u_1}{2}}_{\in U} \in U \cap W$$

$$\Rightarrow \quad \text{baza în } U \cap W : \left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \right\}$$

$$\dim_{\mathbb{R}} U \cap W = 1$$

$$\underbrace{u_1, \frac{u_2}{2} - \frac{u_1}{2}, w_2 - \frac{u_2}{2} + \frac{u_1}{2}}_{\text{baza în } U + W} \in U + W$$

$\dim_{\mathbb{R}}(U+W) = 3 \qquad U+W \subseteq \mathbb{R}^3$

$\Rightarrow$ base di $U+W$: $\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow U+W = \mathbb{R}^3$